

Tema 7

Criptologia

Generarea cheii, criptarea si decriptarea unui text

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Command Prompt
Microsoft Windows [Version 10.0.22631.4317]
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C:\Users\rotar>openssl version
OpenSSL 3.4.0 22 Oct 2024 (Library: OpenSSL 3.4.0 22 Oct 2024)

C:\Users\rotar>openssl version -a
version: Extra option: "ûa"
version: Use -help for summary.

C:\Users\rotar>openssl req -new -key yourdomain.key -out yourdomain.csr
Could not open file or uri for loading private key from yourdomain.key: No such file or directory

C:\Users\rotar>openssl version -a
OpenSSL3.4.0 22 Oct 2024 (Library: OpenSSL 3.4.0 22 Oct 2024)
built on: Tue Oct 22 23:27:41 2024 UTC
platform: VC-WIN64A
options: bn(64,64)
compiler: cl /Z7 /Fdssl_static.pdb /Gs0 /GF /Gy /MD /W3 /wd4090 /nologo /O2
-DL_ENDIAN -DOPENSSL_PIC -D"OPENSSL_BUILDING_OPENSSL" -D"OPENSSL_SYS_WIN32" -D
"WIN32_LEAN_AND_MEAN" -D"UNICODE" -D"_UNICODE" -D"_CRT_SECURE_NO_DEPRECATED" -D
"_WINSOCK_DEPRECATED_NO_WARNINGS" -D"NDEBUG" -D_WINSOCK_DEPRECATED_NO_WARNINGS
-D_WIN32_WINNT=0x0502
OPENSSLDIR: "C:\Program Files\Common Files\SSL"
ENGINESDIR: "C:\Program Files\OpenSSL\lib\engines-3"
MODULESDIR: "C:\Program Files\OpenSSL\lib\openssl-modules"
Seeding source: os-specific
CPUINFO: OPENSSL_ia32cap=0xfffff38bfcffff:0x184007a4239c27a9

C:\Users\rotar>openssl req -new -key yourdomain.key -out yourdomain.csr
Could not open file or uri for loading private key from yourdomain.key: No such file or directory

C:\Users\rotar>openssl genrsa -out yourdomain.key 2048

C:\Users\rotar>openssl req -new -key yourdomain.key -out yourdomain.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
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C:\Users\rotar>openssl req -text -in yourdomain.csr -noout -verify
Certificate request self-signature verify OK
Certificate Request:
  Data:
    Version: 1 (0x0)
    Subject: C=RO, ST=Bucharest, L=Andreea, O=IT, OU=yourdomain.com, CN=yourdomain.com, emailAddress=rotaru.andreea.2001@gmail.com
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (2048 bit)
      Modulus:
        00:a5:74:b6:cf:ce:4c:af:f5:a2:51:a9:79:97:31:
        9c:0f:39:ea:da:c3:0d:f1:54:68:d1:d9:78:d6:45:
        38:d9:c3:b5:f2:4f:c0:fc:13:53:93:c4:2d:a4:3e:
        7d:71:64:5f:2c:29:fc:e7:8a:51:3b:ba:c0:e2:57:
        5b:27:38:db:8a:f3:61:db:a3:e6:ac:11:f6:bc:f5:
        0f:a4:36:97:49:c4:bf:08:93:26:3e:9e:ba:74:16:
        45:c0:96:08:7d:59:3c:7b:55:f2:7c:c9:be:fb:9b:
        77:dc:ac:a0:5b:78:c9:9e:7a:da:84:54:5b:33:a2:
        85:63:d6:ff:0c:b0:e4:d0:4b:95:c0:cf:f8:0f:b3:
        95:b3:77:ab:77:68:5e:fa:7b:f8:fd:3a:43:00:26:
        f3:12:13:e0:5c:2f:65:44:8c:4c:55:05:ca:5d:98:
        54:78:62:02:62:38:dc:20:ce:51:22:33:ee:6b:6b:
        b1:7f:c9:7a:c2:68:fa:c3:68:95:b7:00:d8:3b:fd:
        4f:bc:7c:a0:4c:28:85:91:fa:69:5b:c5:88:63:75:
        60:99:ec:86:46:fe:84:8d:52:a2:3b:66:fc:85:2e:
        28:38:3a:34:ce:56:20:a3:a2:91:fa:61:b0:cd:4e:
        a3:cc:74:6d:49:97:f9:d7:74:b5:c1:4d:5e:52:e0:
        1a:2f
      Exponent: 65537 (0x10001)

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C:\Users\rotar>openssl genrsa -out private_key.pem 2048

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C:\Users\rotar>openssl rsa -in private_key.pem -text -noout
Private-Key: (2048 bit, 2 primes)

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modulus:
  00:b4:dd:cc:7a:a2:1c:fd:69:6c:d5:d3:30:04:c7:
  f7:ad:26:fc:7f:f8:51:f6:c6:30:35:a5:4f:35:3e:
  55:a4:67:58:c5:3b:16:ae:ce:15:cb:f8:b8:fd:a3:
  b6:34:16:d2:59:7c:3d:8d:ca:20:24:e3:9f:62:78:
  29:0b:4f:03:c0:d7:26:45:ec:ea:90:97:bd:6b:d4:
  ad:ed:32:a9:f0:3d:c4:4f:f1:a2:df:55:16:61:a3:
  a5:08:24:87:c7:1a:44:f3:bd:6c:4b:73:6d:5d:e3:
  d4:34:cb:2b:76:19:d8:d2:9a:55:79:53:f5:3a:38:
  f5:76:14:5c:fb:2e:1e:7b:e5:57:86:d0:76:c6:f5:
  26:eb:3d:17:9a:b1:5a:5b:cd:60:dd:c0:fc:7f:72:
  00:44:29:f9:40:37:03:1f:3f:d9:b4:30:5e:f5:65:
  9f:af:7f:fc:4a:c6:f2:58:79:73:88:7d:9a:68:b0:
  2b:af:ae:fe:e6:41:96:6c:e3:1d:ce:40:82:5c:bd:
  6a:8b:30:bf:ae:82:01:86:bc:00:ef:0b:04:b4:df:
  9e:8e:b8:b7:0e:09:da:f5:9c:a2:d5:2f:61:d6:f6:
  5e:fd:1a:8d:f3:3b:13:6f:b5:da:ba:bc:a5:f8:b0:
  49:d8:07:96:61:30:60:b7:18:73:36:dd:16:8b:65:
  ea:bb

```

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C:\Users\rotar>openssl rsa -in private_key.pem -outform PEM -pubout -out public_key.pem
writing RSA key

```

```

C:\Users\rotar>dir
Volume in drive C is Windows
Volume Serial Number is 12DB-9C15

Directory of C:\Users\rotar

13/11/2024  20:33    <DIR>          .
12/03/2024  16:42    <DIR>          ..
26/08/2024  12:09                5 .bash_history
03/10/2024  00:12    <DIR>          .VirtualBox
30/07/2024  12:10    <DIR>          CLionProjects
12/03/2024  16:24    <DIR>          Contacts
13/11/2024  14:19    <DIR>          Desktop
26/08/2024  11:50    <DIR>          Documents
13/11/2024  19:53    <DIR>          Downloads
12/03/2024  16:24    <DIR>          Favorites
12/03/2024  16:24    <DIR>          Links
12/03/2024  16:24    <DIR>          Music
13/11/2024  19:30    <DIR>          OneDrive
02/09/2024  20:18    <DIR>          Pictures
13/11/2024  20:33                42 plaintext.txt
13/11/2024  20:25            1,732 private_key.pem
26/08/2024  12:25    <DIR>          Projects
13/11/2024  20:25            460 public_key.pem
12/03/2024  16:24    <DIR>          Saved Games
12/03/2024  16:54    <DIR>          Searches
03/09/2024  16:34    <DIR>          Videos
02/10/2024  19:43    <DIR>          VirtualBox VMs
13/11/2024  20:23            1,138 yourdomain.csr
13/11/2024  20:16            1,736 yourdomain.key
               6 File(s)              5,113 bytes
              18 Dir(s)  307,719,786,496 bytes free

```

```

C:\Users\rotar>openssl rsautl -encrypt -inkey public_key.pem -pubin -in plaint
ext.txt -out encrypted.txt
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.

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```

C:\Users\rotar>openssl rsautl -decrypt -inkey private_key.pem -in encrypted.tx
t -out decrypted.txt
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.

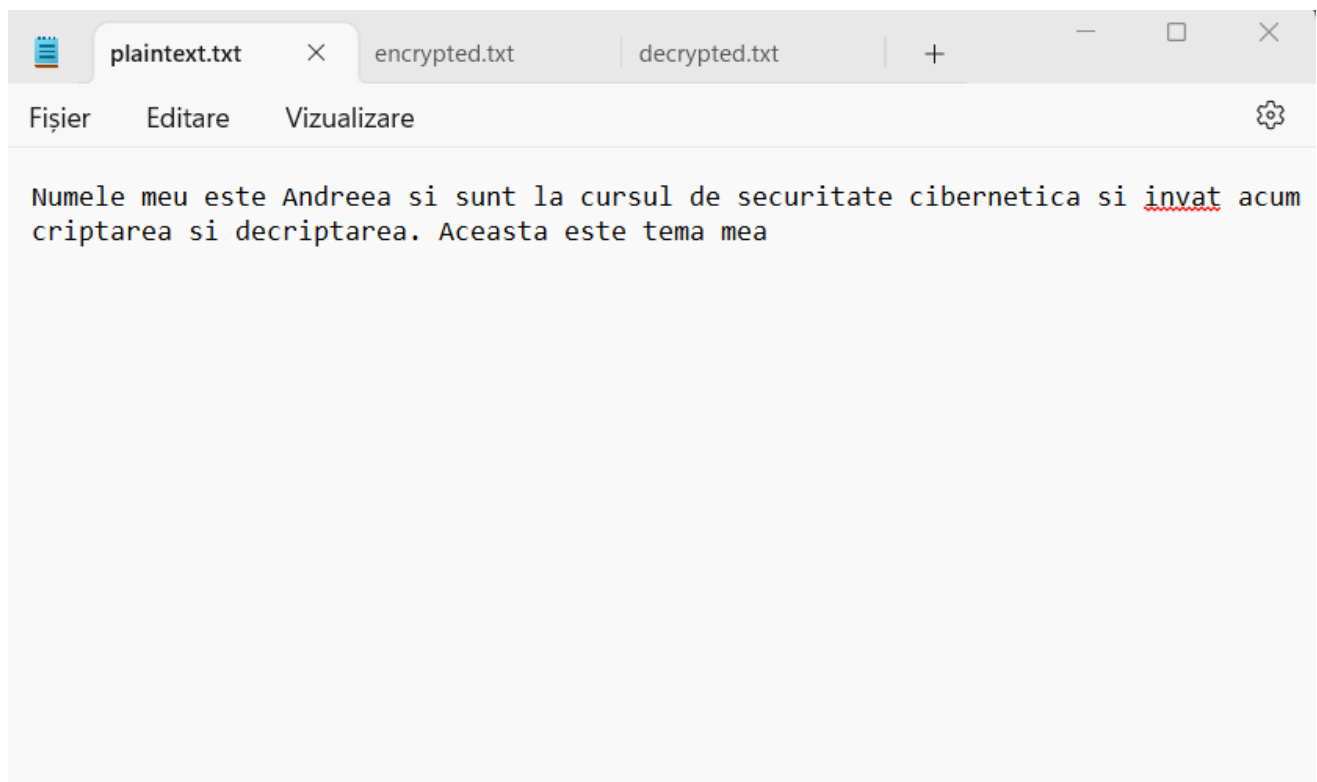
```

Directory of C:\Users\rotar

```

13/11/2024 20:37 <DIR> .
12/03/2024 16:42 <DIR> ..
26/08/2024 12:09 5 .bash_history
03/10/2024 00:12 <DIR> .VirtualBox
30/07/2024 12:10 <DIR> CLionProjects
12/03/2024 16:24 <DIR> Contacts
13/11/2024 20:37 42 decrypted.txt
13/11/2024 14:19 <DIR> Desktop
26/08/2024 11:50 <DIR> Documents
13/11/2024 19:53 <DIR> Downloads
13/11/2024 20:36 256 encrypted.txt
12/03/2024 16:24 <DIR> Favorites
12/03/2024 16:24 <DIR> Links
12/03/2024 16:24 <DIR> Music
13/11/2024 19:30 <DIR> OneDrive
02/09/2024 20:18 <DIR> Pictures
13/11/2024 20:33 42 plaintext.txt
13/11/2024 20:25 1,732 private_key.pem
26/08/2024 12:25 <DIR> Projects
13/11/2024 20:25 460 public_key.pem
12/03/2024 16:24 <DIR> Saved Games
12/03/2024 16:54 <DIR> Searches
03/09/2024 16:34 <DIR> Videos
02/10/2024 19:43 <DIR> VirtualBox VMs
13/11/2024 20:23 1,138 yourdomain.csr
13/11/2024 20:16 1,736 yourdomain.key
      8 File(s)      5,411 bytes
     18 Dir(s)  307,717,922,816 bytes free

```





plaintext.txt

encrypted.txt



decrypted.txt



Fişier

Editare

Vizualizare



0D 畚穰稭回悃罨坏口 백回甞[〓]糲巴回晒爇回器饒口詠回駟(回磧H+샘^々レ訾鯪㉡擊回錐陪섯倬爛回音鰥㉢□□裨諝
蛸選鋅昨畵^々蜥之僂回畱回각回闊鬱➤鴛딕口ㄷm顙攤回≡羈餹敖𣎵▣回演汶阙回聯翬할터回ㅁ㊦경“口뽳X砌
豐口簍e^ㄴ回畀'ᆫx.縶3^ㄴ縶^ㄴ榛^ㄴ



plaintext.txt

encrypted.txt



decrypted.txt



Fişier

Editare

Vizualizare



Numele meu este Andreea si sunt la cursul de securitate cibernetica si invat acum criptarea si decriptarea. Aceasta este tema mea